Sustained Undifferentiated Wide Complex Tachycardia: Adult and Pediatric
C-9 Effective: June 1, 2017

Adult symptomatic tachycardia usually occurs at rates of 120 -150 and typically ≥ 150 beats per minute. Patients symptomatic with heart rates < 150 likely have impaired cardiac function such as CHF.

Most important goal is to differentiate if STABLE or UNSTABLE. A STABLE adult patient may complain of palpitations, chest discomfort, respiratory distress. UNSTABLE adult patient may experience B/P < 90 with signs of shock / poor perfusion and/or a decreasing level of consciousness.

In the event of cocaine/amphetamine/TCA intoxication, refer to the Suspected Overdose Protocol.

Monomorphic VT is likely to be regular and similar in QRS configuration. The QRS will be ≧ 0.12 seconds. Polymorphic VT is irregular and the QRS morphology varies. It is more likely to be unstable and may deteriorate quickly.
Hypotension can be defined as a systolic blood pressure of less than the 5th percentile in a child/infant.

Infant less than 1 month	60 mmHg systolic
1 month to 1 year	70 mmHg systolic
1 year to 10 years	70 plus 2 X age in years systolic
➢ 10 years	90 mm Hg systolic

Begin Initial Medical Care
Obtain a 12 lead ECG and transmit if patient is stable.

A 12 Lead ECG will aid in the determination of VT. The following favor VT.

* Extreme RAD and an upright V1
  * V1 Morphology: 1) Upright complex with taller left rabbit ear 2) Upright complex with single notch or slurring of the R wave
  * V1 morphology: 1) Negative complex with > 0.04 R wave 2) Negative complex with slurring or notch on Q or S wave
* Any negative complex in V6
* RAD with a negative V1
* Negative concordance in V leads
* RS interval in any V lead > 0.10

STABLE ADULT PATIENT:

Contact medical control for treatment.

- If Amiodarone is ordered "AT MD DISCRETION ONLY", administer **Amiodarone 150 mg** in 100 ml D5W over ten minutes. More rapid infusion is not advised. Observe patient closely for hypotension and/or bradycardia.

- If lidocaine is ordered, "AT MD DISCRETION ONLY" administer **Lidocaine 1mg/Kg IVP** slowly at a rate of 50 mg/min.

Treat any initial Amiodarone induced hypotension by stopping infusion and IVF bolus unless contraindicated.
Treat symptomatic bradycardia ≤ rate of 50 with transcutaneous pacing initially. If no results and HR remains ≤ 50 and B/P < 90 administer Dopamine.
Initiate a dopamine infusion IV piggyback at 5 mcg/kg/min. It is best to start a second IV if possible. Use premixed 800 mg of dopamine in 500 cc D5W (creating a 1600 mcg/ml concentration) and infuse at 5-20 mcg/kg/min.

DOPAMINE (Inotropic) Noted number is Microdrops per minute or ml/min

Patient weight in Kg

mcg/Kg/min	2.5	5	10	20	30	40	50	60	70	80	90	100	110	120	130	150
5 mcg		1	2	4	6	8	10	12	14	16	18	20	22	24	26	30
7 mcg		1	2	5	8	11	14	16	19	22	25	28	30	33	36	42
10 mcg	1	2	4	8	11	15	19	23	26	30	34	38	44	48	52	60
15 mcg	1.4	3	6	11	17	23	28	34	39	45	51	56	66	72	78	90
20 mcg	2	4	8	15	23	30	38	45	53	60	68	75	88	96		

Formula: mcg/Kg/min ÷ 25 = microdrops/minute
Contraindications: tachycardic dysrhythmias.
SE: tachycardic dysrhythmias, VT, VF, HTN, N & V, ischemia, AMI; extravasation causes tissue necrosis.

STABLE PEDIATRIC PATIENT:

Ensure adequate oxygenation and ventilation.

Monitor patient and notify Medical Control.

UNSTABLE ADULT PATIENT: Patient with monomorphic VT with decreased LOC and/or a B/P < 90 systolic with signs of shock and or poor perfusion.

Adult Pain Management for Synchronized Cardioversion:
DO NOT DELAY LIFE-SAVING CARE IN ORDER TO SEDATE THE SEVERELY COMPROMISED PATIENT.
If sedation needed:
Versed: **10** mg midazolam IN (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP if GCS ≥ 14

ATTEMPT SYNCHRONIZED CARDIOVERSION WITHIN THE FOLLOWING RANGES:
100 J, 200 J, 300 J, 360 J
Assess LOC, pulses and B/P between each shock delivery to confirm need to escalate dosing.

UNSTABLE ADULT PATIENT: Patient with polymorphic VT with decreased LOC and/or B/P < 90 systolic with signs of shock or poor perfusion.

DEFIBRILATE AS IN VF/PULSELESS VT protocol

UNSTABLE PEDIATRIC PATIENT: Patient with monomorphic VT demonstrates a decreased LOC, poor perfusion **and** hypotensive as defined by a blood pressure below the 5th percentile.

- Synchronize cardiovert at 0.5-1 J/kg, repeated at 2 J/kg if needed.
- **Pediatric Pain Management for Synchronized Cardioversion:** **AT MD DISCRETION ONLY**

UNSTABLE PEDIATRIC PATIENT: Patient with polymorphic VT with a decreased LOC, poor perfusion **and** hypotensive as defined by a blood pressure below the 5th percentile.

DEFIBRILATE AS IN VF/ PULSELESS VT protocol

COMPARING VENTRICULAR ECTOPY WITH ABERRATION

	Left Ventricular Ectopy	Right Bundle Aberration
A	V1 or MCL1 — qR OR R OR R	rSR'
	WITH	WITH
B	V6 or MCL6 — rS OR QS OR R Taller left "rabbit ear" often seen	qRs

	Right Ventricular Ectopy	Left Bundle Aberration
C	V1 or MCL1 — rS (often > 0.03) OR QS (NADIR > 0.06 ON DOWNSTROKE)	QS OR rS
	WITH	WITH
D	V6 or MCL6 — R OR QS (NADIR > 0.07 IN MCL6)	R OR R

RV ectopy and left bundle aberration CAN look identical.

MARRIOTT

Step 1.

Look at the wide QRS beats in MCL1 (V1). If the QRS is UPRIGHT, compare it to the complexes in Row A. If the QRS is DOWNWARD, compare it to the complexes in Row C. When you have determined which complex most closely matches the one you are diagnosing, look above the appropriate row for the diagnosis.

Step 2.

To confirm your diagnosis, look to MCL6. If the beats in MCL1 were upright, compare the MCL6 complexes to those in Row B. If the MCL1 beats were downward, compare the MCL6 complexes to those in Row D.